

Quiz 2**Student Name:** _____**Points:** 30**A Number:** _____**Duration:** 20 Minutes

1. Choose what is true about `nn.Module`

- (a) `nn.Module` is a function used in PyTorch to train a neural network.
- (b) `nn.Module` is an abstract class used to implement a Neural Network Layer.
- (c) `nn.Module` is a data type for Neural Networks in PyTorch
- (d) `nn.Module` is a neural network parameter.

2. Neural network weights in PyTorch can be of type:

- (a) `nn.Parameter`
- (b) `nn.Module`
- (c) `nn.Conv2D`
- (d) `nn.Weight`

3. Hooks in PyTorch are

- (a) functions
- (b) variables
- (c) tensors
- (d) classes

4. What is the difference between parameters and buffers in the realm of PyTorch `nn.Module`?

5. What happens if you don't register a member variable tensor as parameters in PyTorch `nn.Module`?

6. When are forward hooks executed?

- (a) The hook are called every time after `backward()`.
- (b) The hook are called just before calling `forward()`.
- (c) The hook are called every time after weights are updated..
- (d) The hook are called every time after `forward()`.

7. Is it possible to implement a neural network in PyTorch without inhering from `nn.Module`?

- (a) Yes
- (b) No

8. Consider a Network definition

```

1 class Network(nn.Module):
2     def __init__(self):
3         super().__init__()
4         self.layer1 = nn.Linear(10, 20)
5         self.layer2 = nn.Linear(20, 5)
6     def forward(self, x):
7         return self.layer2(self.layer1(x))

```

How many trainable parameters are there in the above network (including biases)?

- (a) 325
 - (b) 300
 - (c) 55
 - (d) 400
9. Imagine a dataset consisting of PNG images with four channels: RGBA. You decided to use a fully connected neural network for creating an image classifier. There 10000 images in your training dataset. Each image has dimension 200×200 . How many features will be there in the input feature matrix to the neural network?

- (a) 10000
- (b) 16000000000
- (c) 40000
- (d) 400000000

10. Minpooling downsamples the image data.

- (a) True
- (b) False

11. An example of order-5 Tensor:

12. Consider a convolution layer definition

```
1 conv = nn.Conv2d(in_channels=4,
2                   out_channels=6,
3                   kernel_size=3,
4                   stride=2,
5                   padding=1)
```

What is the dimension of an example tensor that can serve as an input to the above convolution layer?

- (a) `torch.Size([2, 1, 5, 6])`
- (b) `torch.Size([2, 6, 32, 32])`
- (c) `torch.Size([2, 3, 5, 6])`
- (d) `torch.Size([2, 4, 32, 32])`

13. For the convolution operation in Q12, how many filters will be learnt upon training?

- (a) 4
- (b) 3
- (c) 6
- (d) 2

14. If you are performing regression with a neural network, what should be the final layer in a neural network?

- (a) A linear layer

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- (b) A sigmoid layer
 - (c) A relu layer
 - (d) A tanh layer
15. Applying max pooling with a kernel size of 3×3 to the image of size 5×5 with a stride of 2, the resulting output has size
- (a) 2×2
 - (b) 3×3
 - (c) 5×5
 - (d) 3×5